CAPSTONE PROJECT – DESIGN DOCUMENT

1. Project Information

Project ID: PS-103

Team ID: T050

Project Title: Online Tutoring Marketplace – Session Matching, Cancellation and Tutor Payout Engine

Domain: EdTech – Online Tutoring Marketplace

Programming Language: Python 3.10+

Team Size: 2 Members

2. Project Objective

The objective of this project is to develop a Python-based online tutoring management engine.

The system will automatically match students with suitable tutors based on subject, grade, language, availability, rating and reliability.

It will also manage session bookings, cancellations, rescheduling, pricing, discounts, tutor ratings, and weekly tutor payouts.

The system will operate using input command files without requiring a website or graphical interface.

3. Technologies Used (Planned)

- Python 3.10+ – Programming language
- Visual Studio Code – Code editor
- Git – Version control
- GitHub – Project repository
- CSV files – Input data storage
- pytest – Automated testing

4. Project Scope

The system will support:

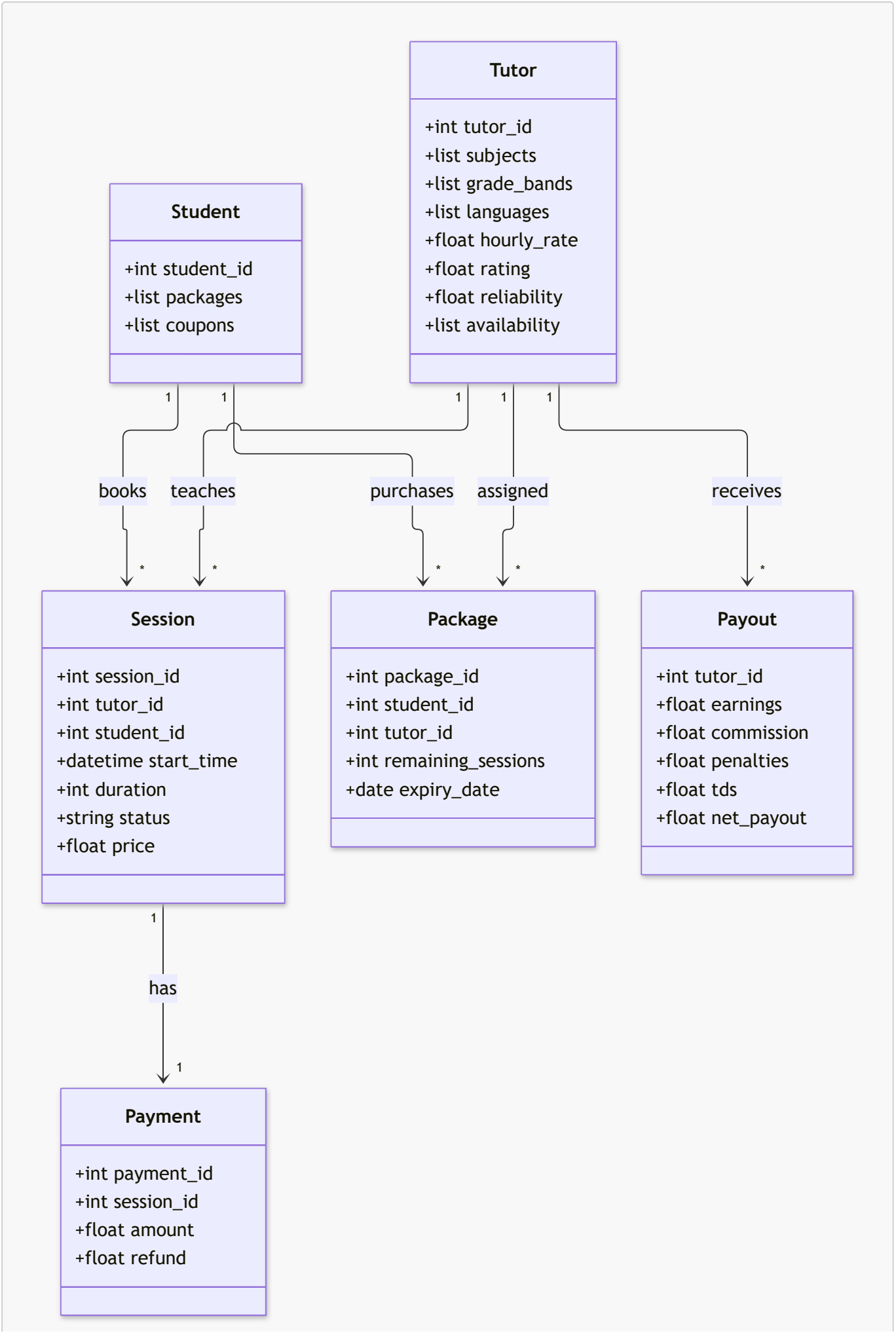
1. Tutor registration and availability
2. Student session requests and tutor matching
3. Session confirmation and rescheduling
4. Student and tutor cancellations
5. Session pricing and discounts
6. Weekly tutor payouts
7. Tutor ratings and reliability
8. Session and payout reports

5. Project Glossary

Term	Meaning
Tutor	A teacher who provides online classes
Student	A person who books a tutoring session
Session	One scheduled online tuition class
Session Request	A student's request for a class
Tutor Matching	Selecting the best suitable tutor
Availability Slot	Time when a tutor is free to teach
Booking	A confirmed tutoring session
Rescheduling	Changing a session to another time
Cancellation	Cancelling a booked session
No-show	Student or tutor failing to attend
Hourly Rate	Tutor's charge for 60 minutes
Surge Pricing	Extra charge during busy hours
Package	A prepaid group of 8 or 16 sessions
Discount	Reduction in the session price
Coupon	Credit applied to a future session
Platform Fee	Amount charged by the marketplace
Commission	Percentage deducted from tutor earnings
Payout	Money payable to a tutor
TDS	Tax Deducted at Source
Reliability Score	Score representing tutor dependability
Rating	Student feedback score from 1 to 5
Demotion	Temporary exclusion due to low ratings
Suspension	Temporary restriction due to repeated late cancellations
Business Rule	A condition the system must follow
Edge Case	A special situation used to test correctness
Session Ledger	Record of sessions and related transactions

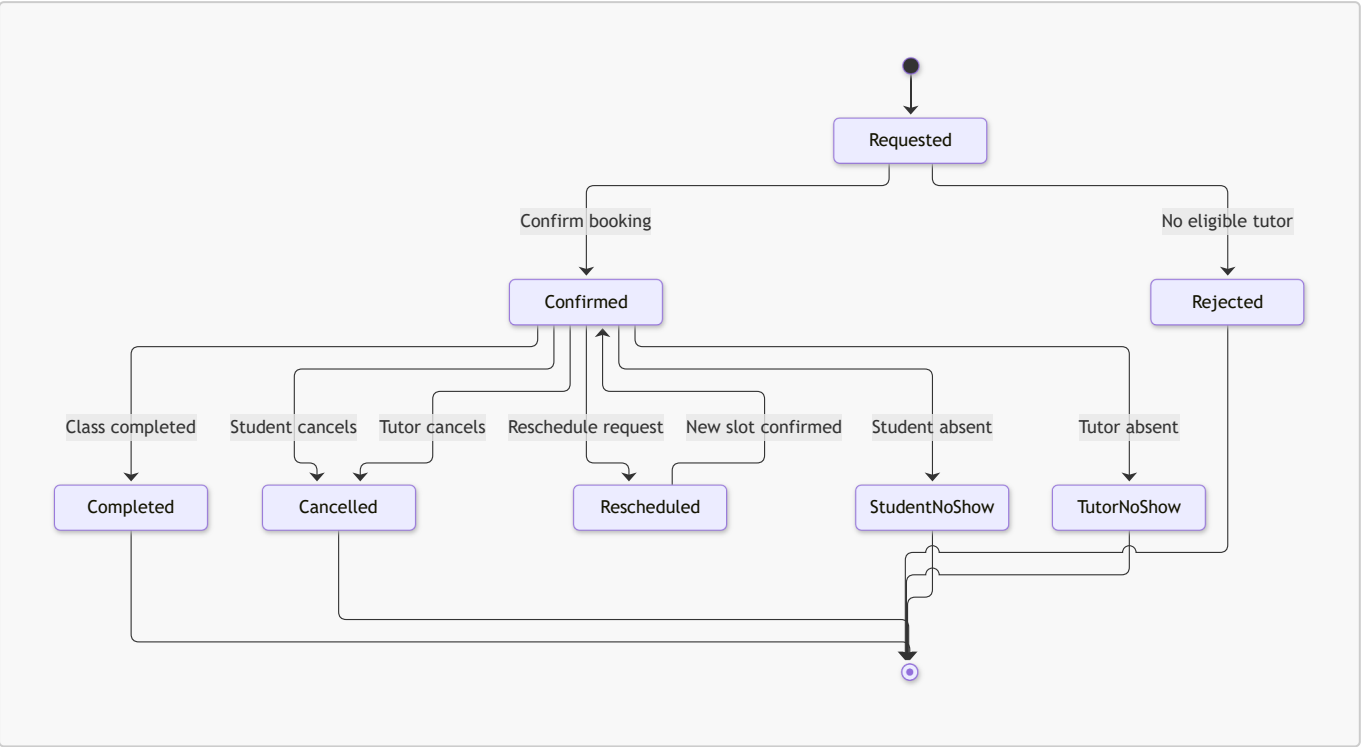
6. Class Diagram

The following diagram shows the main classes and their relationships in the tutoring system.


Tutor
+int tutor_id
+list subjects
+list grade_bands
+list languages
+float hourly_rate
+float rating
+float reliability
+list availability
Student
+int student_id
+list packages
+list coupons
1
1
1
1
1
books
teaches
purchases
assigned
receives
*
*
*
*
*
Session
+int session_id
+int tutor_id
+int student_id
+datetime start_time
+int duration
+string status
+float price
Package
+int package_id
+int student_id
+int tutor_id
+int remaining_sessions
+date expiry_date
Payout
+int tutor_id
+float earnings
+float commission
+float penalties
+float tds
+float net_payout
1
has
1
Payment
+int payment_id
+int session_id
+float amount
+float refund

7. Session State Diagram

This diagram shows the different stages of a tutoring session, from request to final outcome.



State Descriptions

- Requested: Student asks for a tuition session.
- Confirmed: A tutor is assigned and booking is accepted.
- Completed: The tutoring session finishes.
- Cancelled: Student or tutor cancels the booking.
- Rescheduled: The booking moves to another time.
- Rejected: No suitable tutor is available.
- StudentNoShow: Student does not attend the class.
- TutorNoShow: Tutor does not attend the class.

Note: A reschedule may instead cancel the old booking and create a new one, depending on BR-05.

8. Business Rules Register

BR-01: Tutor Eligibility

- Match subject, grade band, language and availability.
- Grade bands: 1-5, 6-8, 9-10 and 11-12.
- Allowed session durations: 30, 45 and 60 minutes.
- Do not assign a tutor with a conflicting booking.

BR-02: Tutor Ranking

- Score = rating * 10 + reliability * 5 - (hourly_rate - cheapest_eligible_rate) / 100

- Select the tutor with the highest score.
- If scores tie, prefer fewer weekly sessions.
- If still tied, select the lower tutor ID.

BR-03: Surge Pricing

- Apply 1.25x pricing on weekdays between 18:00 and 21:00 when fewer than 3 tutors are eligible.
- Do not apply surge pricing to package sessions.

BR-04: Session Packages

- 8-session package: 10% discount.
- 16-session package: 15% discount.
- Package sessions must use the same tutor.
- Packages expire after 60 days.
- Apply the package discount per session.

BR-05: Student Cancellation and Rescheduling

- More than 24 hours before class: 100% refund.
- From 6 to 24 hours before class: 50% refund.
- Less than 6 hours before class: no refund.
- Free rescheduling requires more than 12 hours' notice, the same tutor and a new slot within 7 days.
- Allow a maximum of 2 reschedules per session.

BR-06: Tutor Cancellation

- More than 24 hours: reliability decreases by 0.1.
- From 6 to 24 hours: reliability decreases by 0.3, with a Rs. 200 penalty.
- Less than 6 hours: reliability decreases by 0.5, with a Rs. 500 penalty and a 20% coupon for the student's next session.
- Keep reliability between 0 and 5.
- Increase reliability by 0.05 after completion.

BR-07: No-Show Handling

- Student no-show: charge the full session amount after the tutor has waited 10 minutes.
- Tutor no-show: treat as a late tutor cancellation and issue a full student refund.

BR-08: Weekly Platform Commission

- Fewer than 10 completed sessions: 25%.
- 10 to 24 completed sessions: 20%.
- 25 or more completed sessions: 15%.
- Apply the final weekly tier to all completed sessions in that week.

BR-09: Tutor Payout

- Payout week runs from Monday to Sunday.
- Process weekly payouts on Wednesday.

- Net amount = earnings - commission - penalties.
- Deduct 10% TDS once annual cumulative payout crosses Rs. 30,000, including the crossing payout.
- Minimum payout is Rs. 500.
- Carry amounts below Rs. 500 into the next week.

BR-10: Tutor Rating and Demotion

- Calculate average rating from the latest 20 ratings.
- If average falls below 3.5 after at least 10 ratings, demote the tutor from automatic matching.
- Restore matching eligibility after 5 completed direct-booking sessions.

BR-11: Tutor Suspension

- If a tutor has 3 late cancellations within 30 days, suspend the tutor from automatic matching for 14 days.
- Previously confirmed sessions must still be honored.

BR-12: Rule Precedence and Price Limits

- First check suspension and demotion.
- Then check tutor eligibility.
- Finally rank the eligible tutors.
- Pricing order: package discount, surge exclusion, then coupon discount.
- Apply no more than one coupon per session.
- Minimum final session price is Rs. 100.

9. Mandatory Edge Cases

EC-01: Equal Tutor Scores

If two tutors have the same matching score and the same number of weekly sessions, select the tutor with the lower tutor ID.

Related Rule: BR-02

EC-02: Surge Pricing at 20:30

A student requests a session at 20:30 on a weekday, and exactly 3 tutors are eligible.

Expected Result: Do not apply surge pricing because surge requires fewer than 3 eligible tutors.

Related Rule: BR-03

EC-03: Rescheduling at Exactly 12 Hours

A student requests rescheduling exactly 12 hours before the session.

Expected Result: Reject free rescheduling because the rule requires MORE than 12 hours' notice.

Related Rule: BR-05

EC-04: Annual Payout Crosses Rs. 30,000

A tutor's annual cumulative payout crosses Rs. 30,000 during a weekly payout calculation.

Expected Result: Apply the required 10% TDS to the crossing payout.

Related Rule: BR-09

EC-05: Package Expiry

A package expires before a session booked using that package is scheduled to take place.

Expected Result: Check package validity according to the official package-expiry rule and sample-data specification. Do not silently extend the package.

Related Rule: BR-04

EC-06: Tenth Rating Causes Demotion

A tutor receives their 10th rating, and the average rating falls below 3.5.

Expected Result: Demote the tutor from automatic matching.

Related Rule: BR-10

EC-07: Rs. 499 Payout Carry-Forward

A tutor earns a net payout of Rs. 499 for one week.

Expected Result: Do not release the payout. Carry Rs. 499 forward to the next week. Calculate the next week's commission tier using only that week's completed sessions, then include the carried amount.

Related Rules: BR-08 and BR-09

EC-08: Coupon on a Rs. 110 Session

A student uses a coupon on a 30-minute session whose price before the coupon is Rs. 110.

Expected Result: Apply the eligible coupon but never reduce the final session price below Rs. 100.

Related Rule: BR-12